ORIGINAL ARTICLE

Atezolizumab Plus Chemotherapy for First-Line Treatment of Nonsquamous NSCLC: Results From the Randomized Phase 3 IMpower132 Trial

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ABSTRACT

Introduction: We report the final results of the phase 3 IMpower132 study evaluating atezolizumab plus carboplatin or cisplatin plus pemetrexed (APP) in patients with nonsquamous NSCLC.

Methods: Chemotherapy-naive patients with stage IV nonsquamous NSCLC without sensitizing *EGFR* or *ALK* genetic alterations were randomized in a one-to-one ratio to receive four or six cycles of carboplatin or cisplatin plus pemetrexed (PP) or APP every 3 weeks, followed by maintenance therapy with atezolizumab plus pemetrexed or pemetrexed alone. Co-primary end points were overall survival (OS) and investigator-assessed progression-free survival (PFS).

Results: The intention-to-treat population included 578 patients (APP, n = 292; PP, n = 286). At the primary PFS analysis (May 22, 2018; median follow-up, 14.8 mo), APP exhibited significant PFS improvement versus PP (median = 7.6 versus 5.2 mo, stratified hazard ratio [HR] = 0.60, 95% confidence interval [CI]: 0.49–0.72, $p < 0.0001$). OS for the APP group was numerically better but not statistically significant at the interim (May 22, 2018; median = 18.1 versus 13.6 mo, stratified HR = 0.81, 95% CI: 0.64–1.03, $p = 0.0797$) and final analyses (July 18, 2019; median = 17.5 versus 13.6 mo; stratified HR = 0.86, 95% CI: 0.71–1.06, $p = 0.1546$). The OS and PFS results favored APP versus PP across subgroups. Grade 3 or 4 treatment-related adverse events occurred in 54.6% (APP) and 40.1% (PP) of patients; grade 5 treatment-related events occurred in 3.8% and 2.9%, respectively.

Conclusions: IMpower132 met its co-primary PFS end point but not its co-primary OS end point, with numerical improvement for OS in the APP arm. APP had a manageable safety profile, with no new or unexpected safety signals identified.

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Keywords: Lung cancer; Atezolizumab; Nonsquamous; Non–small cell

Introduction

Lung cancer is a leading cause of cancer-related deaths[1] despite recent advances providing additional treatment options. Approximately 85% of patients with lung cancer present with NSCLC and more than half of NSCLCs have nonsquamous histology.[2] For patients whose tumors do not harbor known driver mutations in the *EGFR* or *ALK* loci, first-line treatment options include immune checkpoint inhibitor monotherapy for those with high expression of programmed death-ligand 1 (PD-L1) and chemoimmunotherapy combinations with or without bevacizumab.[3-6] Atezolizumab is an anti–PD-L1 monoclonal antibody (Tecentriq; Genentech, Inc./F. Hoffmann-La Roche Ltd., South San Francisco, CA) that selectively inhibits PD-L1 interactions with programmed cell death protein 1 (PD-1) and B7.1, thereby reinvigorating and enhancing anticancer immunity and potentially preserving immune homeostasis by leaving the PD-L2/PD-1 interaction intact.[7,8] In the United States, atezolizumab is approved as first-line monotherapy in patients with chemotherapy-naive metastatic squamous or nonsquamous NSCLC and in combination regimens for first-line treatment of patients with metastatic nonsquamous NSCLC and no *EGFR* or *ALK* genomic tumor aberrations.[9-12] Patients with metastatic nonsquamous NSCLC without *EGFR* mutation or *ALK* aberration may receive atezolizumab in combination with nab-paclitaxel and carboplatin. In the European Union, atezolizumab is approved in combination with bevacizumab, paclitaxel, and carboplatin for first-line treatment of patients with metastatic nonsquamous NSCLC.[13]

Pemetrexed (Alimta; Eli Lilly, Indianapolis, IN) is widely used in platinum-based chemotherapy combinations owing to its favorable toxicity profile.[14] Tumor cell (TC) death by platinum-based chemotherapy is believed to expose the immune system to high levels of tumor antigens, and invigorating tumor-specific T-cell immunity by inhibiting PD-L1 signaling may improve treatment response.[15,16] Promising results from a Phase Ib study (NCT01633970) of platinum-based doublet chemotherapy plus atezolizumab supported the rationale for further evaluation in patients with NSCLC.[17,18] The phase 3 KEYNOTE-189 study revealed improved progression-free survival (PFS) and overall survival (OS) with the addition of the PD-1 inhibitor pembrolizumab to platinum-based pemetrexed combination chemotherapy compared with chemotherapy alone for first-line treatment of advanced nonsquamous NSCLC.[19]

We report here the PFS and OS findings from the phase 3 IMpower132 study (NCT02657434) investigating the efficacy and safety of adding atezolizumab to

carboplatin or cisplatin plus pemetrexed for patients with nonsquamous NSCLC and no sensitizing *EGFR* or *ALK* driver mutations.

Materials and Methods

Trial Design

IMpower132 is a randomized, phase 3, multicenter, open-label study designed to evaluate the safety and efficacy of atezolizumab in combination with cisplatin or carboplatin plus pemetrexed (APP) compared with cisplatin or carboplatin plus pemetrexed (PP) in chemotherapy-naive patients with stage IV nonsquamous NSCLC. Conducted at cancer centers globally, IMpower132 fully adhered to the guidelines on Good Clinical Practice and the Declaration of Helsinki. Written informed consent was obtained from all patients. F. Hoffmann-La Roche Ltd./Genentech, Inc., sponsored the IMpower132 study, provided study drugs, and collaborated with academic authors with regard to study design, data collection, analysis, and interpretation. An independent data monitoring committee reviewed safety throughout the study. Protocol approval was obtained from independent ethics committees for each site. All authors verified that the study was conducted per the protocol and vouch for data accuracy and completeness. Manuscript drafts were prepared by the authors with editorial assistance from professional medical writers funded by the sponsor. All authors approved the submission of this manuscript.

Study Participants

Eligible patients were adults older than or equal to 18 years of age with signed written informed consent and had histologically or cytologically confirmed stage IV nonsquamous NSCLC with measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, an Eastern Cooperative Oncology Group performance status of 0 or 1, and no previous treatment for metastatic disease. Histologically or cytologically confirmed stage IV nonsquamous NSCLC was determined using the seventh edition of the American Joint Commission on Cancer staging system.[20] A treatment-free interval of greater than or equal to 6 months was required if patients had previous chemotherapy or radiotherapy with curative intent for the nonmetastatic disease. Pretreatment tumor tissue was submitted if available; however, patients were still eligible if tumor tissue was depleted or otherwise not available. Patients with untreated central nervous system metastases, autoimmune disease, or tumors harboring sensitizing mutations in *EGFR* gene or *ALK* genetic alterations (as assessed at a local or central laboratory) or who had received previous immunotherapy were excluded.

Interventions

Patients were randomized on a one-to-one ratio to APP or PP regimens. Induction treatment consisted of four to six 21-day cycles (number of cycles at the investigator's discretion). Atezolizumab was administered at 1200-mg fixed dose every 21 days. Cisplatin 75 mg/m^2 or carboplatin at a target area under the concentration-time curve of 6 mg/mL/min was administered every 21 days. Pemetrexed 500 mg/m^2 was administered every 21 days. All study drugs were administered by intravenous infusion. Maintenance therapy with atezolizumab plus pemetrexed (APP arm) or pemetrexed (PP arm) was given every 21 days after induction therapy until disease progression, unacceptable toxicity, or death. Patients in the APP arm could continue atezolizumab after progression if there was evidence of clinical benefit (as assessed by the investigator). No crossover between treatment groups was allowed.

Outcome Assessments

The co-primary end points were investigator-assessed PFS (measured per RECIST 1.1) and OS in the intention-to-treat (ITT) population. Other objectives included investigator-assessed objective response rate (ORR), investigator-assessed duration of response (DOR), and safety.

Tumor tissue collection for the assessment of PD-L1 expression was not required for study enrollment. For patients who submitted tissue for the biomarker assessment (59.5% of the ITT population), baseline PD-L1 tumor expression was centrally evaluated in archival tissue or tissue obtained from a biopsy at screening using the Ventana SP142 PD-L1 immunohistochemistry (IHC) assay (Ventana Medical Systems, Inc., Tucson, AZ). The scoring algorithm of the SP142 IHC assay measures expression of PD-L1 on both TCs and tumor-infiltrating immune cells (ICs).[21] The PD-L1–high group comprised patients with PD-L1 expression in greater than or equal to 50% of TCs or greater than or equal to 10% of tumor-infiltrating ICs (TC3 or IC3). The PD-L1–low group had PD-L1 expression in greater than or equal to 1% and less than 50% of TCs, or greater than or equal to 1% and less than 10% of tumor-infiltrating ICs (TC1/2 or IC1/2). The PD-L1–negative group (TC0 and IC0) had PD-L1 expression in less than 1% of TCs and less than 1% of tumor-infiltrating ICs. Patients underwent tumor assessments at baseline and every 6 weeks after cycle 1 day 1 for the first 48 weeks and every 9 weeks thereafter until radiographic disease progression per RECIST 1.1, loss of clinical benefit (for patients in APP arm who continued on atezolizumab after radiographic disease progression per RECIST 1.1), or death. The incidence,

nature, and severity of adverse events (AEs) were assessed using the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.0. Changes in vital signs, physical findings, and clinical laboratory results were also monitored during and after treatment administrations.

Sample Size Calculation and Randomization Procedures

IMpower132 was designed to enroll approximately 568 patients on the basis of the number of events required to illustrate a statistically significant difference in both PFS and OS between APP and PP. The overall type I error rate for treatment comparisons in the primary end points was controlled using the group sequential Holm procedure for the one-sided test at 0.025 (Supplementary Fig. 1).[22] PFS in the ITT population was tested at a one-sided α level of 0.002 and OS in the ITT population was tested at a one-sided α level of 0.023. If only PFS was significant, then OS would be tested at a one-sided α level of 0.025; otherwise, OS would be tested at the initially assigned one-sided α level of 0.023. If only OS was significant, PFS would be tested at a one-sided α level of 0.025; otherwise, PFS would be tested at the initially assigned one-sided α level of 0.002. The number of events required to reveal efficacy in the ITT population with regard to PFS was estimated, assuming a one-sided significance level of 0.002 with 96% power to detect a hazard ratio (HR) of 0.65, corresponding to an improvement in median PFS from 6 months to 9.2 months. The number of events required to exhibit efficacy in the ITT population with regard to OS was estimated, assuming a one-sided significance level of 0.023 with 81% power to detect an HR of 0.75, corresponding to an improvement in median OS from 14 months to 18.7 months.

Patient information was collected by means of a central interactive voice/Web response system. Permuted-block randomization with a block size of four was used to allocate patients in a one-to-one ratio to each treatment group. Randomization was stratified by sex, smoking status (never versus current or former), Eastern Cooperative Oncology Group performance status (0 versus 1), and chemotherapy regimen (cisplatin versus carboplatin).

Statistical Methods

Primary efficacy analyses were conducted in the ITT population, which consisted of all randomized patients whether or not assigned treatment was received; the safety population consisted of patients who received any amount of study treatment. The primary PFS analysis was planned to be conducted when approximately 458 PFS events had occurred in the ITT population and 10 or more months after the last patient was enrolled in the study, whichever occurred later. An interim OS analysis was planned to be conducted at the same time as the primary PFS analysis. The final OS analysis was planned to be conducted when approximately 398 OS events had occurred. The Lan-DeMets approximation to the Pocock boundary was used to compute the interim and final OS analysis stopping boundaries. PFS and OS were compared between the APP and PP arms using a stratified log-rank test; the HR was estimated using a stratified Cox regression model. Kaplan-Meier methodology was used to estimate median PFS and OS. The Brookmeyer-Crowley method was used to construct the 95% CI for median PFS and OS. The PFS rate at 6-month and 1-year landmark time points and the OS rate at 1- and 2-year landmark time points were estimated using Kaplan-Meier methodology. Descriptive statistics were used to evaluate patient demographics and safety outcomes. Statistics were calculated using the Statistical Analysis System version 9.0 or higher, or R version 3.5.1.

Results

Participants

Between April 7, 2016, and May 31, 2017, a total of 578 patients were enrolled from 164 study sites across 26 countries in North America, Western Europe, Eastern Europe, Latin America, and Asia. The ITT population consisted of 578 patients randomized to receive APP (n = 292) or PP (n = 286), of whom 565 received study treatment and composed the safety population (Fig. 1). At the primary PFS and interim OS analysis (data cutoff, May 22, 2018), 264 patients (45.7%) were still on the study—147 (50.3%) from the APP arm and 117 (40.9%) from the PP arm. The median follow-up was 14.8 months, and the minimum follow-up was 11.7 months. At the time of the final OS analysis (data cutoff, July 18, 2019), 168 patients (29.1%) were still on the study: 91 (31.2%) from the APP arm and 77 (26.9%) from the PP arm. The median follow-up was 28.4 months, and the minimum follow-up was 25.5 months. Baseline demographic and clinical characteristics were similar between treatment arms (Table 1).

Progression-Free Survival

At the time of primary PFS analysis (data cutoff, May 22, 2018), PFS events per investigator assessment had occurred in 209 patients (71.6%) in the APP arm and 249 patients (87.1%) in the PP arm. The APP treatment arm exhibited statistically significant PFS improvement compared with the PP arm (median = 7.6 versus 5.2 mo; stratified HR = 0.60, 95% confidence interval [CI]: 0.49–0.72, $p < 0.0001$) (Fig. 2*A*). Landmark PFS rates in the

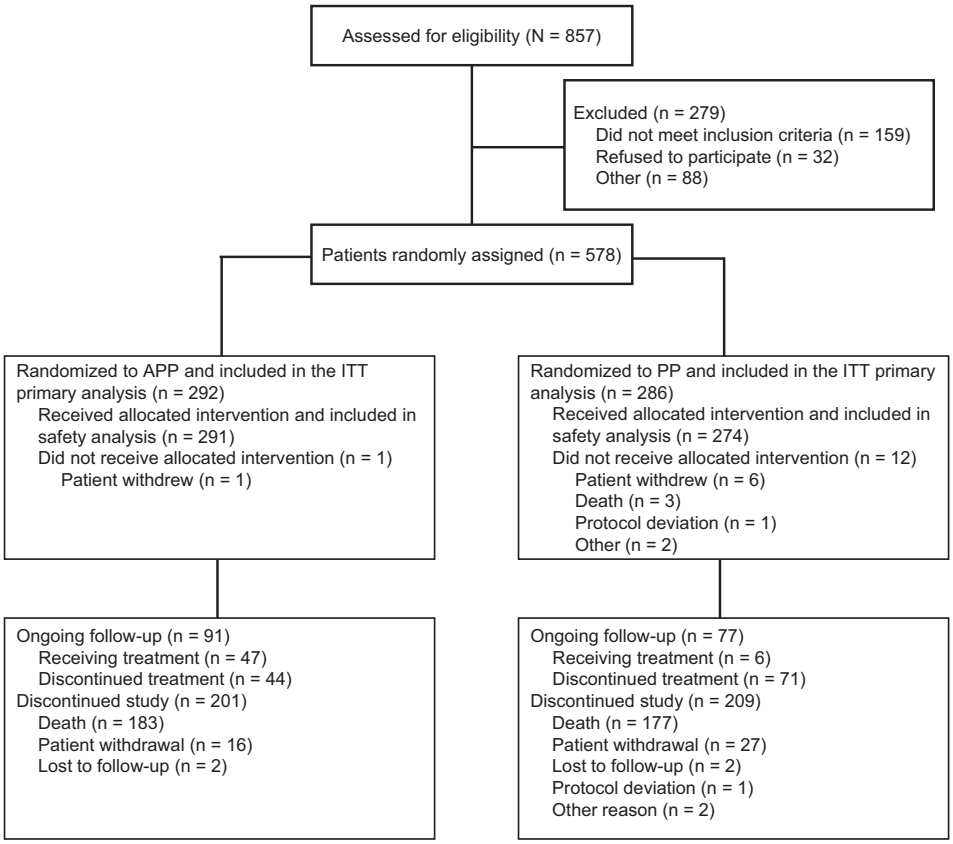


Figure 1. Disposition of patients in IMpower132. APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; ITT, intention-to-treat; PP, carboplatin or cisplatin plus pemetrexed.

APP and PP arms were 59.1% and 40.9% at 6 months and 33.7% and 17.0% at 12 months, respectively. Observed PFS differences across key patient subgroups were generally consistent with the primary analysis, including the PFS benefit among Asian patients (median = 10.2 versus 5.3 mo with APP and PP, respectively; HR = 0.42, 95% CI: 0.28–0.63) (Fig. 2*C*). The exploratory analysis of PFS by PD-L1 expression was performed in the biomarker-assessable population (approximately 59.5% of the ITT population who had available tissue at enrollment). In this biomarker-assessable subgroup, median PFS in the high PD-L1 expression group (TC3 or IC3, 13% of the biomarker-assessable population) for patients receiving APP was 10.8 months versus 6.5 months for those receiving PP (unstratified HR = 0.46, 95% CI: 0.22–0.96) (Fig. 2*B*). In the negative (TC0 and IC0) and low (TC1/2 or IC1/2) PD-L1 expression groups, median PFS for APP versus PP was 8.5 versus 4.9 months (unstratified HR = 0.45, 95% CI: 0.31–0.64) and 6.2 versus 5.7 months (unstratified HR = 0.80, 95% CI: 0.56–1.16), respectively (Fig. 2*B*).

The updated PFS at final analysis (data cutoff, July 18, 2019) was 7.7 and 5.2 months for the APP and PP arms, respectively (HR = 0.56, 95% CI: 0.47–0.67) (Supplementary Fig. 2). PFS events per investigator assessment had occurred in 238 patients (81.5%) in the APP arm and 274 patients (95.8%) in the PP arm. Updated PFS rates in the APP and PP arms at final analysis were 34.4% and 17.3% at 12 months and 19.2% and 4.4% at 24 months, respectively.

Overall Survival

As of the interim OS analysis (data cutoff, May 22, 2018), 137 patients (46.9%) in the APP arm and 154 patients (53.8%) in the PP arm had died, reflecting a non-statistically significant difference between the APP and PP arms (median = 18.1 versus 13.6 mo; stratified HR = 0.81, 95% CI: 0.64–1.03, $p = 0.0797$). As part of the exploratory analysis, PD-L1 expression subgroups were analyzed in the biomarker-assessable population. No clear trends emerged by level of PD-L1 expression (APP versus PP median, unstratified HR [95% CI]: high PD-L1 subgroup [TC3 or IC3, 13% of the biomarker-assessable population], median OS was not reached for either arm, 0.41 [0.13–1.25]; low PD-L1 subgroup [TC1/2 or IC1/2], 12.7 versus 16.2 mo, 1.14 [0.73–1.80]; PD-L1–negative subgroup [TC0 and IC0], 19.9 versus 10.5 mo, 0.71 [0.47–1.07]). Interim OS findings were generally consistent across patient subgroups, including in Asian patients (stratified HR = 0.68 [95% CI: 0.37–1.24]).

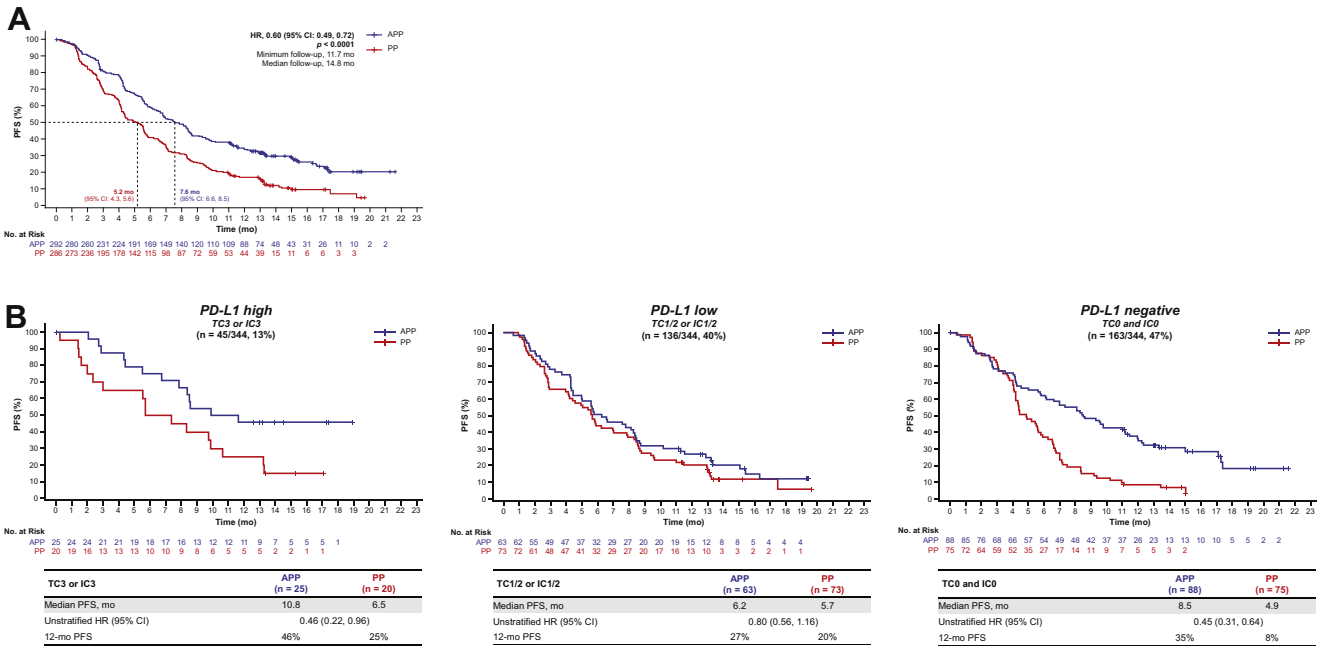
Table 1. Baseline Demographic and Clinical Characteristics

Characteristics	APP (n = 292)	PP (n = 286)	All Patients (N = 578)
Female sex, n (%)	100 (34.2)	94 (32.9)	194 (33.6)
Age, y			
Median (range)	64.0 (31-85)	63.0 (33-83)	63.5 (31-85)
≥ 65, n (%)	139 (47.6)	118 (41.3)	257 (44.5)
Race, n (%)			
White	193 (66.1)	203 (71.0)	396 (68.5)
Black or African American	2 (0.7)	4 (1.4)	6 (1.0)
Asian	71 (24.3)	65 (22.7)	136 (23.5)
American Indian or Alaska Native	1 (0.3)	1 (0.3)	2 (0.3)
Unknown or undisclosed	25 (8.6)	13 (4.5)	38 (6.6)
Ethnicity, n (%)			
Hispanic or Latino	17 (5.8)	21 (7.3)	38 (6.6)
Not Hispanic or Latino	243 (83.2)	241 (84.3)	484 (83.7)
Not reported	27 (9.2)	20 (7.0)	47 (8.1)
Unknown	5 (1.7)	4 (1.4)	9 (1.6)
Weight, kg[a]			
Median (range)	69.2 (38.5-128.0)	69.5 (39.0-127.0)	69.4 (38.5-128.0)
Tobacco use history, n (%)			
Never	37 (12.7)	30 (10.5)	67 (11.6)
Current or former	255 (87.3)	256 (89.5)	511 (88.4)
ECOG performance status, n (%)			
0	126 (43.2)	114 (40.1)[b]	240 (41.7)
1	166 (56.8)	170 (59.9)	336 (58.3)
Liver metastasis, n (%)	37 (12.7)	36 (12.6)	73 (12.6)
4 induction cycles, n (%)	197 (67.5)	190 (66.4)	387 (67.0)
PD-L1 expression, n (%)[c]	n = 176	n = 168	n = 344
TC3 or IC3	25 (14.2)	20 (11.9)	45 (13.1)
TC1/2 or IC1/2	63 (35.8)	73 (43.5)	136 (39.5)
TC0 and IC0	88 (50.0)	75 (44.6)	163 (47.4)
EGFR mutation status, n (%)			
Positive	2 (0.7)	5 (1.7)	7 (1.2)
Negative	290 (99.3)	281 (98.3)	571 (98.8)
KRAS mutation status, n (%)			
Positive	19 (6.5)	20 (7.0)	39 (6.7)
Negative	50 (17.1)	47 (16.4)	97 (16.8)
Unknown	223 (76.4)	219 (76.6)	442 (76.5)

APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; ECOG, Eastern Cooperative Oncology Group; IC, tumor-infiltrating immune cell; PD-L1, programmed death-ligand 1; PP, carboplatin or cisplatin plus pemetrexed; TC, tumor cell.
[a]Weight was available for 290 patients in the APP arm and 274 patients in the PP arm.
[b]ECOG performance status scores were available for 284 patients in the PP arm.
[c]Prevalence was calculated in the biomarker-assessable population (n = 344).

As of the final OS analysis (data cutoff, July 18, 2019), 192 patients (65.8%) in the APP arm and 197 patients (68.9%) in the PP arm had died. The difference in OS between the APP and PP treatment arms was not statistically significant (median, 17.5 versus 13.6 mo; stratified HR = 0.86 [95% CI: 0.71–1.06]; $p = 0.1546$) (Fig. 3*A*). The OS rates at 12 months in the APP and PP arms were 59.7% and 55.0%, respectively. At 24 months, OS rates were 39.1% and 34.0% in the APP and PP arms, respectively. OS in the key patient subgroups was generally similar to the overall patient population (Fig. 3*C*) including the Asian subgroup (HR = 0.73, 95% CI: 0.46–1.15). In patients with high PD-L1 expression (TC3 or IC3, 13% of biomarker-assessable population), the median was not reached in the APP arm (n = 25) compared with 26.9 months in the PP arm (n = 20; HR = 0.73, 95% CI: 0.31–1.73; Fig. 3*B*). In the low PD-L1 expression subgroup (TC 1/2 or IC 1/2; 40% of biomarker-assessable population), median OS was 12.7 versus 16.2 months in the APP arm (n = 63) and PP arm (n = 73), respectively (HR = 1.18, 95% CI: 0.80–1.76) (Fig. 3*B*). In the PD-L1–negative subgroup (TC0 and IC0; 47% of biomarker-assessable population), the median OS was 15.9 months in the APP arm (n = 88) versus 10.5 months in the PP arm (n = 75; HR = 0.67, 95% CI: 0.46, 0.96) (Fig. 3*B*).

As of the updated OS analysis (data cutoff, July 18, 2019), more patients in the PP arm received nonprotocol follow-up cancer therapy (n = 164; 57.3%) compared with those in the APP arm (n = 113; 38.7%). The PP arm received a higher proportion of subsequent



TC3 or IC3	APP (n = 25)	PP (n = 20)
Median PFS, mo	10.8	6.5
Unstratified HR (95% CI)	0.46 (0.22, 0.96)	
12-mo PFS	46%	25%

TC1/2 or IC1/2	APP (n = 63)	PP (n = 73)
Median PFS, mo	6.2	5.7
Unstratified HR (95% CI)	0.80 (0.56, 1.16)	
12-mo PFS	27%	20%

TC0 and IC0	APP (n = 88)	PP (n = 75)
Median PFS, mo	8.5	4.9
Unstratified HR (95% CI)	0.45 (0.31, 0.64)	
12-mo PFS	35%	8%

Figure 2. PFS in the IMpower132 primary analysis. PFS *(A)* in the ITT population, *(B)* by PD-L1 expression, and *(C)* in key patient subgroups. Data cutoff was on May 22, 2018. PD-L1-high: PD-L1 expression on greater than or equal to 50% of TC or greater than or equal to 10% of IC; PD-L1-low: greater than or equal to 1% and less than 50% of TC or greater than or equal to 1% and less than 10% of IC; PD-L1-negative: less than 1% of TC and less than 1% of IC. [a]Stratified HR for ITT population; unstratified HR for all other subgroups. APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; CI, confidence interval; ECOG PS, Eastern Cooperative Oncology Group performance status; HR, hazard ratio; IC, tumor-infiltrating immune cells; IHC, immunohistochemistry; ITT, intention-to-treat; PD-L1, programmed death-ligand 1; PFS, progression-free survival; PP, carboplatin or cisplatin plus pemetrexed; TC, tumor cells.

immunotherapy than the APP arm (PP n = 131, 45.8% versus APP n = 16, 5.5%) (Supplementary Table 1). The most common subsequent immunotherapy agents used were nivolumab (APP arm, 3.4%; PP arm, 26.6%), pembrolizumab (APP arm, 0.3%; PP arm, 10.1%), and atezolizumab (APP arm, 1.0%; PP arm, 6.3%). Subsequent chemotherapy was given in 34.9% of patients in the APP arm versus 29.0% in the PP arm.

Objective Response

As the primary PFS analysis (data cutoff, May 22, 2018), confirmed ORR and DOR were favorable for patients receiving APP versus PP overall. Patients receiving APP had a confirmed ORR of 47% (95% CI: 41.1–52.8) compared with 32% (95% CI: 26.8–37.9) among those in the PP arm, with median DOR of 10.1 months (95% CI: 7.2–13.3) and 7.2 months (95% CI: 5.7–9.0), respectively. Analysis of overall response by PD-L1 status was also available for the biomarker-assessable population (Supplementary Table 2).

Safety

As of the July 18, 2019, data cutoff, the median duration of treatment in the APP arm was 6.2 months for atezolizumab (range, 0–36 mo), 5.1 months for pemetrexed (range, 0–34 mo), 2.3 months for carboplatin (range, 0–6 mo) and 2.1 months for cisplatin (range, 0–5 mo). In the PP arm, the median duration of treatment was 3.7 months for pemetrexed (range, 0–31 mo), 2.2 for carboplatin (range, 0–5 mo), and 2.1 months for cisplatin (range, 0–4 mo).

The safety profile of APP was consistent with the known risks associated with the individual treatment components; no new safety signals were observed. Grade 3 or 4 treatment-related AEs were observed in 54.6% of patients receiving APP and 40.1% of patients receiving PP, and grade 5 treatment-related AEs were observed in 3.8% and 2.9% of patients, respectively (Table 2). Overall, immune-mediated AEs were observed in 50.9% and 38.7% of patients receiving APP or PP, respectively. Grade 3 or 4 immune-mediated AEs were observed in 15.1% of patients receiving APP and in 6.9% of patients receiving PP, and Grade 5 immune-mediated AEs in three patients (1%) and one patient (0.4%) from the APP and PP groups, respectively. Discontinuation of any drug owing to an AE occurred in 28.5% of the patients in the APP arm and 18.2% in the PP arm. Discontinuation rates owing to AEs in the APP and PP arms, respectively, were 23.4% and 11.7% for pemetrexed, 3.8% and 4.7% for cisplatin, and 4.5% and 7.3% for carboplatin. The

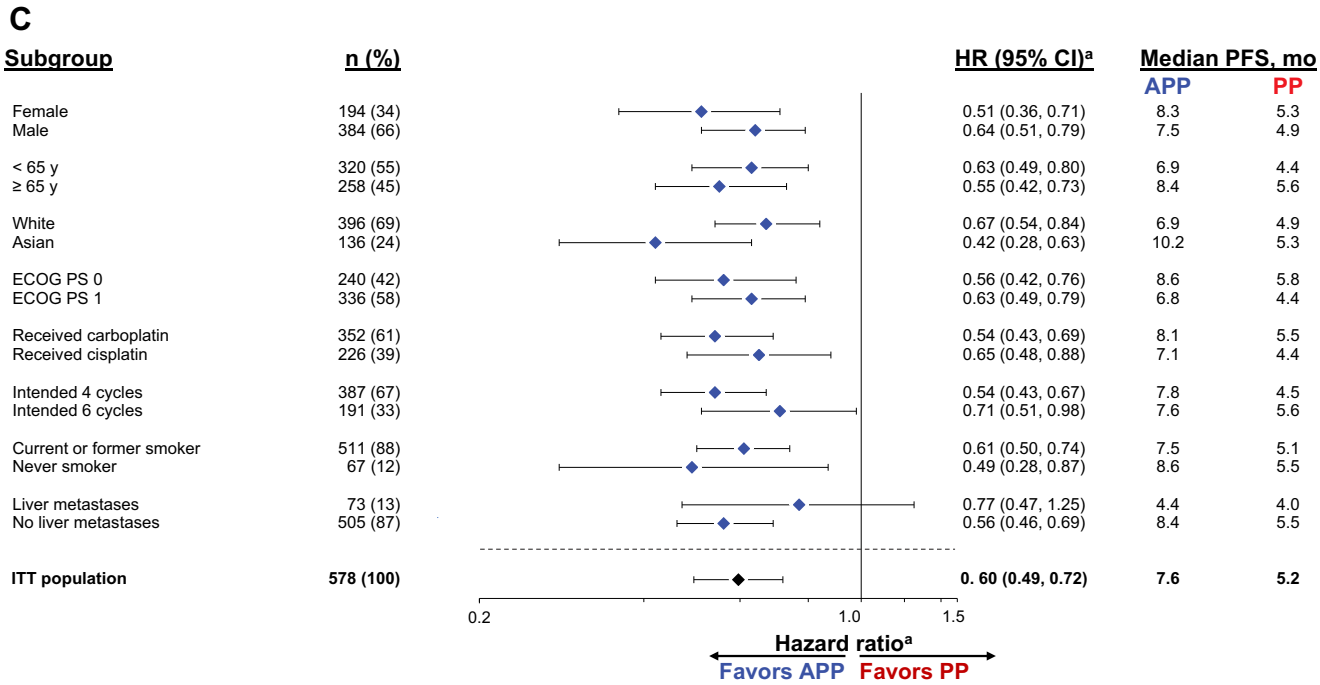


Figure 2. (*continued*).

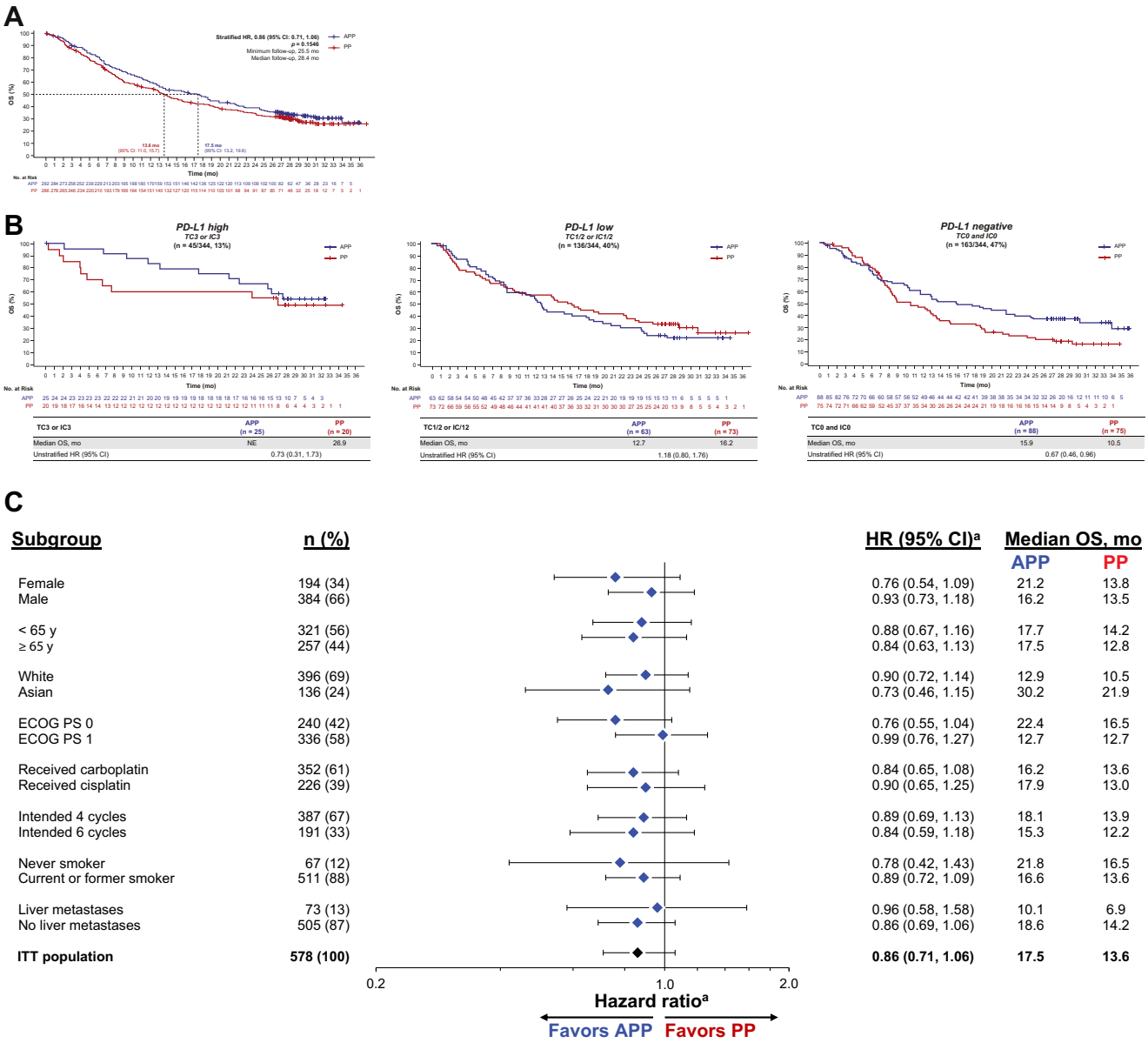
discontinuation rate owing to AEs for atezolizumab was 16.2% in the APP arm. The most common AEs of special interest (any grade, $\geq$ 5% in either arm) summarized by medical concepts for APP and PP, respectively, were rash (25.8% versus 21.5%), hypothyroidism (8.2% versus 2.2%), and pneumonitis (6.2% versus 2.2%) (Table 3).

Discussion

The IMpower132 study revealed significantly longer PFS in patients receiving APP versus PP for first-line treatment of NSCLC, meeting the co-primary PFS end point, but did not cross the significance ($p < 0.0001$) boundary for the co-primary OS end point. Atezolizumab in combination with carboplatin or cisplatin and pemetrexed exhibited a manageable safety profile, with no new or unexpected safety signals.

It should be noted that patients in the APP arm had longer median OS in the interim and final OS analyses than those in the PP arm; however, the differences did not achieve statistical significance. Among patients randomized to the PP arm, 57% and 46% received subsequent second-line and beyond (2L+) therapy and immunotherapy, respectively, and the corresponding rates in patients randomized to APP were only 39% and 6%. The higher incidence of subsequent 2L+ anticancer therapy (57% versus 39%), particularly immunotherapy (46% versus 6%), in the PP arm may have impacted survival outcomes. Patients in the APP arm had a longer median OS in the interim analysis than at the final analysis (18.1 and 17.5 mo, respectively), whereas patients in the PP arm had a median OS of 13.6 months in both analyses. Although only 59.5% of the ITT was part of the biomarker-assessable population, a notable shift was observed from the interim to the final analyses in the subgroup of patients with PD-L1–high tumors (interim OS HR = 0.41, 95% CI: 0.13–1.25 versus final OS HR = 0.73, 95% CI: 0.31–1.73). This observation warrants further analysis but may have been partly attributable to the high rates of subsequent 2L+ treatment—nearly half of patients in the PP arm received subsequent immunotherapy. This was also an open-label trial, which may have impacted the use of subsequent immunotherapy for patients who were randomized to the PP arm.

Exploratory analysis of PD-L1 as a biomarker in this study was limited owing to the availability of archived tumor tissues or fresh biopsies for centralized testing, resulting in a reduced number of biomarker-assessable patients. Thus, the data for PD-L1 subgroups presented in this study should be interpreted accordingly. The reasons that the PD-L1–low subgroup did not exhibit results similar to those of the high or negative subgroups remain an area of further investigation. The small sample size on the basis of available tissue, prognostic factors, or molecular imbalance associated with this specific subgroup may have contributed to this observation. Findings related to PFS and OS were generally consistent across subgroups on the basis of baseline demographic and clinical characteristics, including PFS and OS



TC3 or IC3	APP (n = 25)	PP (n = 20)
Median OS, mo	NE	26.9
Unstratified HR (95% CI)	0.73 (0.31, 1.73)	

TC1/2 or IC/12	APP (n = 63)	PP (n = 73)
Median OS, mo	12.7	16.2
Unstratified HR (95% CI)	1.18 (0.80, 1.76)	

TC0 and IC0	APP (n = 88)	PP (n = 75)
Median OS, mo	15.9	10.5
Unstratified HR (95% CI)	0.67 (0.46, 0.96)	

Figure 3. OS in the IMpower132 final overall survival analysis. OS *(A)* in the ITT population, *(B)* by PD-L1 expression, and *(C)* in key patient subgroups. Data cutoff was on July 18, 2019. PD-L1-high: PD-L1 expression on greater than or equal to 50% of TC or greater than or equal to 10% of IC; PD-L1-low: greater than or equal to 1% and less than 50% of TC or greater than or equal to 1% and less than 10% of IC; PD-L1-negative: less than 1% of TC and less than 1% of IC. [a]Stratified HR for ITT population; unstratified HR for all other subgroups. APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; CI, confidence interval; ECOG PS, Eastern Cooperative Oncology Group performance status; HR, hazard ratio; IC, tumor-infiltrating immune cells; IHC, immunohistochemistry; ITT, intention-to-treat; OS, overall survival; PD-L1, programmed death-ligand 1; PP, carboplatin or cisplatin plus pemetrexed; TC, tumor cells.

benefits observed in the subgroup of Asian patients, suggesting that an additional benefit of atezolizumab plus platinum-based chemotherapy may exist in this population.

Though direct comparisons cannot be made between studies, varying OS and PFS outcomes have been observed across phase 3 studies in NSCLC. In the KEYNOTE-189 study evaluating pembrolizumab with pemetrexed and platinum-based chemotherapy versus pemetrexed and platinum-based chemotherapy alone, the co-primary OS and PFS end points were both met.[19] Despite the inherent challenges of indirectly assessing clinical trial populations, it may be noted that OS was greater in the PP comparator arm of IMpower132 than

Table 2. Summary of Safety Among Treated Patients in IMpower132

Category	APP (n = 291)	PP (n = 274)
Median treatment duration (range), mo		
Atezolizumab	6.2	NA
Pemetrexed	5.1	3.7
Carboplatin	2.3	2.2
Cisplatin	2.1	2.1
All-cause AEs, n (%)	287 (98.6)	266 (97.1)
Grade 3 or 4	185 (63.6)	151 (55.1)
Grade 5	23 (7.9)	15 (5.5)
Treatment-related AEs, n (%)	266 (91.4)	240 (87.6)
Grade 3 or 4	159 (54.6)	110 (40.1)
Grade 5	11 (3.8)	8 (2.9)
Serious AEs, n (%)	146 (50.2)	89 (32.5)
Treatment-related	100 (34.4)	48 (17.5)
AEs leading to withdrawal, n (%)		
Of any treatment	83 (28.5)	50 (18.2)
Of atezolizumab	47 (16.2)	NA
AEs of special interest, n (%)	148 (50.9)	106 (38.7)
Grade 3 or 4	44 (15.1)	19 (6.9)
Grade 5	3 (1.0)	1 (0.4)
AEs of special interest treated with systemic corticosteroids (%)	47 (16.2)	13 (4.7)

AE, adverse event; APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; NA, not applicable; PP, carboplatin or cisplatin plus pemetrexed.

in the comparator arm of the KEYNOTE-189 pembrolizumab trial (median OS = 13.6 mo in IMpower132 PP arm after a median 14.8 mo of follow-up; 11.3 mo in KEYNOTE-189 comparator arm after a median 10.5 mo of follow-up, and 10.7 mo after a median 18.7 mo of follow-up).[19,23] These observations were also reflected in the high PD-L1 subgroups, in which patients in the IMpower132 PP arm had a median OS of 26.9 months, and the KEYNOTE-189 chemotherapy arm had a median OS of 10.1 months in the most recent analysis (median follow-up, 23.1 mo).[24] Improved OS and PFS were seen across the PD-L1 subgroups evaluated in KEYNOTE-189 (>90% had assessable PD-L1 expression).[19] PD-L1 tumor expression was greater than or equal to 50% in 32% to 34% of patients in KEYNOTE-189 treatment arms, and relative benefits of pembrolizumab were most robust in this subgroup. In IMpower150, a PFS benefit was observed across PD-L1

Table 3. AEs of Special Interest

AEs of special interest, n (%)	APP (n = 291)		PP (n = 274)	
	All Grades	Grade 3 or 4	All Grades	Grade 3 or 4
Rash	75 (25.8)	9 (3.1)	59 (21.5)	5 (1.8)
Hypothyroidism	24 (8.2)	1 (0.3)	6 (2.2)	0
Pneumonitis	18 (6.2)	6 (2.1)[a]	6 (2.2)	3 (1.1)[a]
Hepatitis (diagnosis)	13 (4.5)	7 (2.4)[a]	2 (0.7)	0
Infusion-related reactions	7 (2.4)	0	3 (1.1)	1 (0.4)
Hyperthyroidism	7 (2.4)	1 (0.3)	3 (1.1)	0
Meningoencephalitis	4 (1.4)	2 (0.7)	2 (0.7)	0
Pancreatitis	4 (1.4)	1 (0.3)	2 (0.7)	2 (0.7)
Severe cutaneous adverse reactions	4 (1.4)	2 (0.7)	2 (0.7)	0
Nephritis	4 (1.4)	4 (1.4)	1 (0.4)	0
Colitis	4 (1.4)	1 (0.3)	0	0
Diabetes mellitus	3 (1.0)	2 (0.7)	1 (0.4)	1 (0.4)
Adrenal insufficiency	3 (1.0)	0	0	0
Vasculitis	3 (1.0)	1 (0.3)	0	0
Myocarditis	1 (0.3)	1 (0.3)	1 (0.4)	1 (0.4)
Rhabdomyolysis	1 (0.3)	0	0	0
Systemic immune activation	1 (0.3)	1 (0.3)	0	0
Hypophysitis	1 (0.3)	1 (0.3)	0	0

AE, adverse event; APP, atezolizumab plus carboplatin or cisplatin plus pemetrexed; PP, carboplatin or cisplatin plus pemetrexed.
[a]Grade 5 (fatal) events observed.

subgroups in patients who received atezolizumab plus bevacizumab and platinum-based chemotherapy versus bevacizumab and platinum-based chemotherapy, including those with low PD-L1 expression.[12] CheckMate 227 reported an OS benefit with nivolumab plus ipilimumab versus chemotherapy regardless of PD-L1 expression level.[25] The CheckMate 9LA study of nivolumab plus ipilimumab and chemotherapy versus chemotherapy alone in first-line NSCLC revealed a sustained survival benefit that was consistent across PD-L1 and histology subgroups, suggesting potential differences across PD-1 and PD-L1 treatment regimens.[26]

Atezolizumab in combination with carboplatin or cisplatin and pemetrexed exhibited a manageable safety profile. AEs were consistent with the known risks of individual study treatment or the symptoms of the patients' underlying diseases. Overall AEs, including serious AEs, were more frequent in the APP arm, which also had a longer duration of follow-up for safety outcomes. The AE reporting window was longer in the APP owing to a longer median treatment duration in the APP arm versus the PP arm. The proportions of patients with reported Grade 3 or 4 AEs, fatal AEs, serious AEs, and AEs leading to any treatment discontinuation were higher in the APP arm than the PP arm. Reported events were distributed across multiple organs and systems, with no specific trend identified.

Atezolizumab plus pemetrexed and platinum chemotherapy offered longer PFS and a nonsignificant but numerically better OS compared with pemetrexed plus platinum-based chemotherapy in the first-line treatment of patients with nonsquamous NSCLC, with a manageable safety profile. Adding atezolizumab immunotherapy to pemetrexed and platinum-based chemotherapy may offer an improved first-line treatment option for patients with nonsquamous NSCLC.

Data Availability Statement

Qualified researchers may request access to individual patient-level data through the clinical study data request platform (https://vivli.org/). Further details on Roche's criteria for eligible studies are available here (https://vivli.org/members/ourmembers/). For further details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents, see here (https://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm).

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Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org and at https://doi.org/10.1016/j.jtho.2020.11.025.

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